A Comprehensive Review on Document Forgery Detection

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***Abstract*—With the increasing reliance on digital documents, the risk of document forgery has grown significantly. Image forgery detection has traditionally focused on alterations within natural images. However, document image forgery presents unique challenges, as it often involves modifications in text regions, which are more difficult to detect visually. These challenges pose additional obstacles for digital forensic techniques. This paper provides a comprehensive review of document forgery detection. It covers the various types of forgery, key detection challenges, and commonly used evaluation metrics. Furthermore, the review examines seven primary detection methods, analyzing their strengths and limitations. By comparing these methods, this paper offers valuable insights to support ongoing research and help improve practical solutions for safeguarding document authenticity and security."**

1. Introduction

In recent years, advances in image editing technologies [1]–[4] have made digital document manipulation increasingly common. At the same time, there has been a growing reliance on digital media for communication and information sharing. Document forgery, defined as the unauthorized alteration of digital or scanned documents, presents significant challenges in fields such as law, academia, finance, and corporate governance [5]. These manipulations not only undermine document credibility but also pose serious security, ethical, and financial risks. As a result, detecting document forgery has become an essential area of research. This has drawn attention from scholars and industry professionals, who are developing sophisticated methods to identify and prevent forgery attempts. Figure 1 illustrates the growth in publications from 1998 to 2024, as indexed by the Web of Science database under searches for Document Forgery Detection.

Building on previous research [6]–[10], common types of document tampering include copy-move, which duplicates a document section; splicing, which adds external content; and removal, which erases content and fills it with surrounding elements. Additional methods, such as generation of new text or graphics and deep synthesis using AI, create realistic manipulations that further complicate detection.

To assess the effectiveness of forgery detection systems, a range of evaluation metrics is applied. Metrics such as Intersection over Union (IoU), Precision, Recall, F-score (F1), Area Under the Curve (AUC), and the Kappa Coefficient are widely used. Each metric evaluates different aspects of a system's performance, such as accuracy, consistency, and reliability, providing a multi-faceted view of its effectiveness.

Document forgery detection employs a range of methods to ensure document authenticity. Early approaches [11]–[15] rely on spatial analysis, examining shapes and text layouts for inconsistencies that may indicate tampering. Some printer source-based techniques [16]–[20] analyze unique printing characteristics to verify document origins. Additionally, security verification methods [21]–[24], such as watermarking, embed authenticity markers within documents. Visual feature analysis [25]–[28] detects tampering by examining color and texture variations. Hyperspectral and multi-spectral imaging [29]–[37] enhances this by iden-

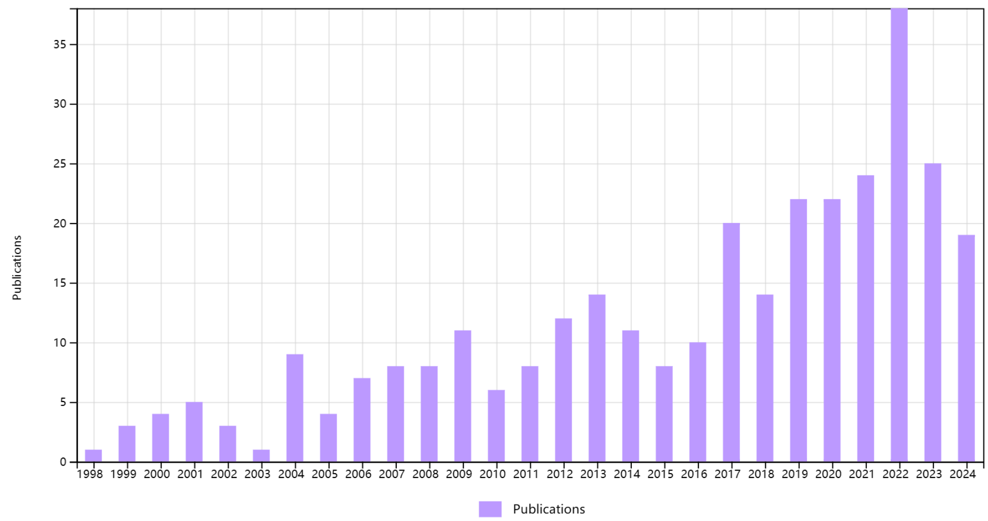

Figure 1. Number of papers published between 1998 and 2024 on *Document Forgery Detection* in the Web of Science database

tifying inconsistencies in ink and material properties. Frequency domain analysis [6], [38], [39] further complements these methods by revealing manipulation artifacts in structural patterns. Finally, deep learning-based methods [7]–[9], [40] provide high accuracy through automated feature detection. Together, these methods form a comprehensive framework for robust document forgery detection.

2. Types and Challenges of Document Forgery

Document forgery detection involves identifying various types of tampering techniques, each with distinct characteristics and detection challenges. The primary types of forgery include copy-move, splicing (or cutting [7]), removal (or erasing [7]), generation, and deep synthesis [6], [8]–[10]. Figure 2 illustrates examples of some forgery type. The following parts provide detailed descriptions of these tampering techniques.

2.1. Copy-Move

Copy-move tampering involves duplicating a region within the same image and relocating it elsewhere [7], often used to obscure sensitive information or to replicate elements misleadingly. Detecting copy-move forgeries is challenging because the duplicated areas are frequently modified slightly (e.g., through rotation, scaling, or subtle color adjustments) to blend with the surroundings, making standard feature-matching techniques less effective.

2.2. Splicing

Splicing (or cutting) entails inserting content from one image into another [6], creating a seamless integration that alters the original context. Splicing is difficult to detect, particularly when the added content closely resembles the color, texture, and lighting of the target image, requiring sophisticated texture and edge analysis techniques to identify discrepancies in blending.

2.3. Removal

Removal (or erasing) involves deleting a region in the document image and using surrounding pixels to fill in the area [7] [8], often done to hide critical information. Detection is complicated because the filled area may appear consistent with the natural structure, making it hard to distinguish from unaltered sections without advanced edge and texture consistency checks.

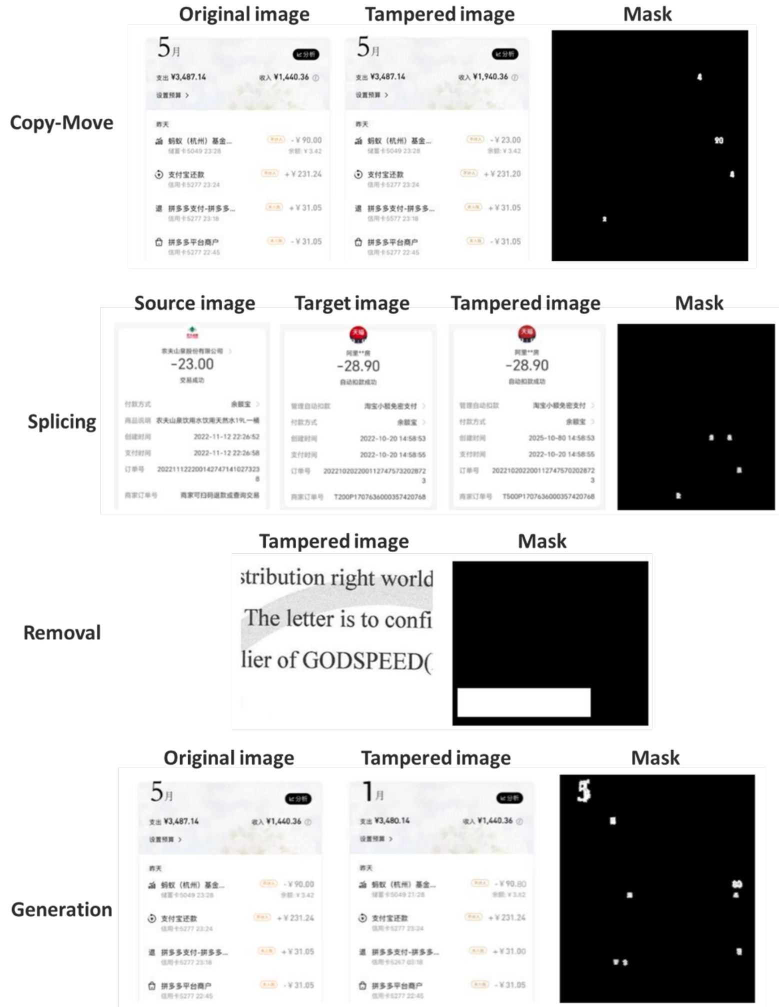

Figure 2. Some forgery types in document forgery

2.4. Generation

Generation involves adding new text or image elements directly to different target regions within a text image. This may include inserting text or images that appear seamlessly integrated into the original document [10]. Detecting generation-based tampering is challenging due to the precision with which new elements can be integrated. Advanced techniques allow added content to closely match the original style, font, and alignment, making it difficult to distinguish authentic content from manipulated regions.

2.5. Deep synthesis

Deep synthesis refers to modifying information in text images through layer-based compositing or AI-generated content (AIGC). This technique often employs multiple layers or deep learning models to alter or create realistic visual elements within the text imag [10]. Identifying deep synthesis is particularly complex because AI-based modifications often lack visible inconsistencies. Sophisticated layer integration and synthetic content can mimic genuine visual features with high fidelity, making traditional detection methods less effective at revealing tampered areas.

3. Evaluation Metrics

The evaluation metrics used in document forgery detection are similar to those in image manipulation detection [41]–[44]. These metrics include Intersection over Union (IoU), Precision, Recall, F-score (F1), Area Under the Curve (AUC), and the Kappa Coefficient. Each metric provides a different perspective on the detection system's accuracy, consistency, and overall efficacy.

3.1. IoU

Intersection over Union (IoU) is a metric that quantifies the overlap between predicted tampered regions and the ground truth [10], [40]. Higher IoU values indicate better overlap, and thus, higher accuracy in identifying tampered areas. IoU is calculated as the ratio of the intersection area to the union area between the predicted and actual regions:

$$\mathrm{IoU} = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FP} + \mathrm{FN}}$$

where True Positive (TP) and True Negative (TN) represent tampered pixels and authentic pixels that have been correctly identified. False Positive (FP) and False Negative (FN), on the other hand, refer to misclassified pixels, with FP indicating authentic pixels incorrectly labeled as tampered, and FN indicating tampered pixels mistakenly identified as authentic. [40]

3.2. Precision

Precision measures the accuracy of the tampered regions detected by the model, reflecting the proportion of true positive results out of all positive predictions made by the model [10]. It is calculated as:

$$\mathrm{Precision} = \frac{TP}{TP + FP}$$

High precision indicates a low false positive rate, which is crucial for reliable detection.

3.3. Recall

Recall, also known as Sensitivity, evaluates the ability of the model to correctly identify all tampered regions within the document image [10]. It is calculated by dividing true positives by the sum of true positives and false negatives:

$$\mathrm{Recall} = \frac{TP}{TP + FN}$$

High recall means that most tampered regions are successfully detected, minimizing missed detections.

3.4. F-score

F-score (F1) is the harmonic mean of Precision and Recall, offering a single metric that balances both false positives and false negatives [10], [40]. It is particularly useful when there is an uneven class distribution. The F-score is given by:

$$\mathrm{F1} = 2 \cdot \frac{Precision \times \mathrm{Recall}}{Precision + \mathrm{Recall}}$$

This score provides a more comprehensive view of model performance when both Precision and Recall are important.

3.5. AUC

Area Under the Curve (AUC) represents the area under the Receiver Operating Characteristic (ROC) curve, which plots the true positive rate against the false positive rate across various threshold values [10]. AUC ranges from 0 to 1, with higher values indicating better performance, as it reflects the model's ability to differentiate between tampered and non-tampered areas effectively.

3.6. Kappa Coefficient

Kappa Coefficient (or Cohen's Kappa) measures the level of agreement between the model's predictions and the ground truth, accounting for agreement that could occur by chance [31]. It is calculated as:

$$\text{Kappa} = \frac{p_o - p_e}{1 - p_e}$$

where p_o is the observed accuracy and p_e is the expected accuracy by chance. A high Kappa value indicates strong agreement, signifying that the model's performance is not due to random guessing.

4. Method

Document forgery detection encompasses a range of techniques targeting various aspects of document integrity. Early spatial feature analysis examines geometric shapes and text layouts for inconsistencies, while printer source-based detection uses machine learning to classify documents based on unique printing characteristics. To enhance authenticity, security verification techniques like watermarking and digital signatures have been implemented. As forgery techniques evolved, visual feature-based methods, including color and texture analysis, emerged to distinguish alterations. Alternative techniques, such as hyperspectral imaging (HSI) and multispectral imaging (MSI), leveraging unique spectral signatures of inks and materials for enhanced detection. Additionally, frequency domain analysis uncovers hidden tampering artifacts through the examination of frequency patterns within document images. Finally, deep learning techniques facilitate automated feature extraction with high accuracy, effectively capturing complex forgery details. The following sections will elaborate on each technique in detail.

4.1. Early Spatial Feature Analysis

Early research on document forgery detection emphasizes the use of spatial features to identify tampering through geometric shapes and text differences. Figure 3 shows some key spatial features used in detection. Shang et al. [11] introduce a method that analyzes distortion mutations in geometric parameters, such as translation and rotation, to detect anomalies in both Chinese and English documents, proving effective even under low JPEG compression. Bertrand et al. [12] employ Conditional Random Fields to classify character fonts, identifying discrepancies in font properties relative to surrounding text, thus enhancing detection capabilities against document manipulation.

Van Beusekom et al. [13] propose an automated approach that analyzes text-line skew angles and alignments, facilitating tampering identification in high-volume environments. Bertrand et al. [14] focus on character-level intrinsic features, using a discriminant feature space to distinguish genuine characters from forgeries, particularly addressing challenges in low-resolution documents. Finally, Zramdini and Ingold [15] present a statistical method for optical font recognition that utilizes global typographical features.

These spatial feature-based techniques demonstrate notable effectiveness in detecting common forgery patterns. However, they may have limitations in detecting more subtle or complex tampering, particularly when forgeries are designed to closely mimic genuine document features.

4.2. Printer Source-Based Detection

Another approach in document forgery detection is printer source identification, which leverages machine learning and feature analysis to classify documents based on unique printing characteristics, aiming to achieve high detection accuracy. Figure 4 shows the generic process for printer source-based detection methods. The following session discuss several studies that utilize printer source-based detection methods.

Bibi et al. [16] adopt a text-independent approach, utilizing deep learning features from pretrained CNNs to enhance classification performance. Shang et al. [17] focus on character-based features, analyzing noise energy and contour roughness to effectively differentiate between laser and inkjet printers. Elkasrawi and Shafait [18] leverage unique noise patterns through a supervised learning framework, addressing the challenges posed by specific scan resolutions. In terms

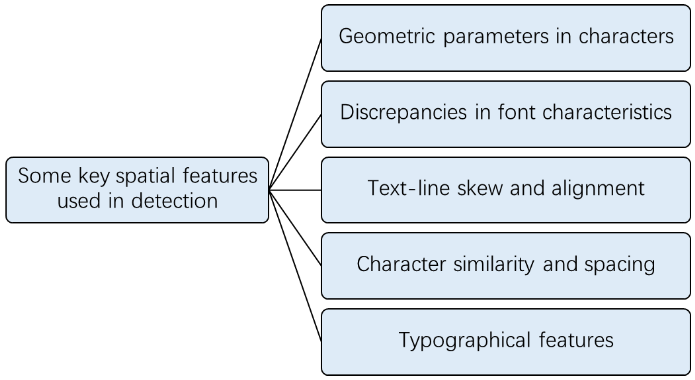

Figure 3. Some key spatial features used in document forgery detection

of feature analysis, Lampert et al. [19] implement a machine learning classification system that analyzes printing techniques at the letter level, providing visualizations that assist non-expert users in interpretation. Meanwhile, Mikkilineni et al. [20] employ texture features derived from gray-level co-occurrence matrices, contributing to forensic document authentication.

Together, these studies illustrate the diverse methodologies employed in printer source identification for document forgery detection, each offering distinct advantages into improving accuracy. However, the effectiveness of these techniques can vary based on the type of forgery and the quality of the documents analyzed, while limitations such as dependency on specific printer models and environmental factors can affect overall reliability.

4.3. Security Verification Techniques

To further ensure document authenticity, some document forgery detection focus on security verification techniques to ensure document integrity and authenticity. Boonkrong [21] presents a cryptographic hash function-based system that detects modifications in academic digital documents, offering a fast and accurate solution that surpasses traditional methods like digital signatures and blockchain. Dlamini et al. [23] propose a comprehensive verification system combining OCR, cryptographic hashing, digital signatures, and 2D barcodes, enhancing robustness against font variations and document damage. Cheddad et al. [24] introduce a self-embedding algorithm that uses a 1D hash combined with 2D inverse Fast Fourier Transform (iFFT) for encryption, providing resilience against noise and compression while allowing access to original documents after manipulation. Wibowo Putro and Luthfi [22] present a method that integrates OCR and perceptual hashing into QR codes, enhancing the security of physical documents against forgery attacks on both text and visual elements.

To sum up, these research shows various methods for verifying document security, with each having its own benefits for ensuring documents' integrity and authenticity. However, their success can depend on the document types and attack methods, and they might require expert knowledge to use and could be vulnerable to advanced forgery techniques.

4.4. Visual Feature-Based Methods

4.4.1. Color and Texture Analysis. Color and texture analysis methods in forgery detection focus on extracting distinct visual characteristics to distinguish genuine from tampered document images. Each technique relies on analyzing color channels or texture patterns to identify irregularities. Gornale et al. [25] applies RGB color channels and GLCM texture features, achieving high classification accuracy using SVM to differentiate between forged and authentic handwritten or printed documents. Megahed et al. [26] employs ink color analysis by combining RGB features

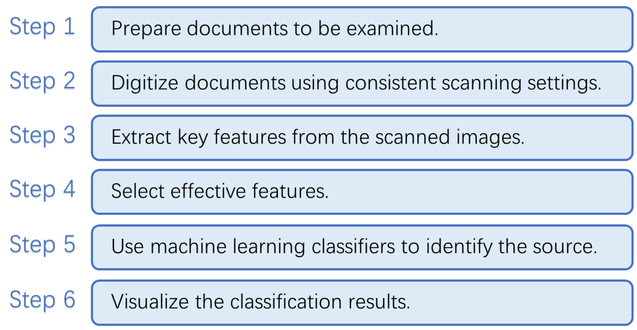

Figure 4. Generic process for printer source-based detection methods

with RMSE and the Thompson Tau method, effectively localizing forged areas with high detection efficiency. Cruz et al. [27] leverages Local Binary Patterns (LBP) to capture textural inconsistencies in documents, integrating contextual information for robust detection; it also introduces a realistic dataset simulating authentic forgery scenarios. Lastly, Gorai et al. [28] uses histogram matching with texture features such as LBP and Gabor filters, offering a simple yet efficient approach that requires minimal prior knowledge. Together, these methods illustrate the potential of color and texture-based approaches in identifying document forgery, each optimized for specific visual markers and document types.

4.4.2. Distortion and Content Consistency-Based Detection. Distortion and content consistency-based forgery detection methods focus on analyzing discrepancies in document structure or print characteristics to accurately locate tampered regions. Each technique relies on detecting inconsistencies in either the spatial or spectral properties of document patterns. Chen et al. [45] proposes a print-scan (PS) distortion model that leverages spatial and spectral distortions in color dot patterns to identify forgery, achieving high localization accuracy by comparing these patterns with a reference template. Ahmed et al. [46] focuses on static template distortions to enhance detection precision, utilizing layout filtering and approximation techniques that reduce runtime while improving speed and accuracy in forgery detection. In contrast, Zhao et al. [47] addresses text forgery in complex backgrounds using a low-cost deep learning algorithm, combining disentanglement, text skeleton guidance, and post-processing to overcome challenges in intricate character structures. This approach successfully circumvents authentication systems, even with minimal target-domain samples. Together, these methods illustrate diverse approaches to forgery detection, each suited to specific scenarios—from template consistency checks to handling complex text alterations in document images.

Overall, while these visual feature-based techniques provide valuable tools for forgery detection, their effectiveness can be contingent on the quality of the input images and the complexity of the forgery techniques employed. Limitations may include sensitivity to noise, reliance on specific document types, and the potential for sophisticated forgers to exploit weaknesses in these methods.

4.5. Hyperspectral and Multispectral Imaging Techniques

As shown in Figure 5, hyperspectral imaging (HSI) and multispectral imaging (MSI) are advanced optical techniques that analyze material composition by capturing spectral information across different wavelength bands. HSI, with its high spectral resolution, captures images over hundreds of narrow bands, making it suitable for identifying subtle differences in inks, pigments, and paper fibers in documents. In contrast, MSI captures fewer and broader bands, making it more efficient for preliminary forensic inspections requiring quicker imaging.

Early works, such as those by Silva et al. [37], introduced near-infrared hyperspectral imaging combined with chemometric methods to detect various types of forgery, demonstrating HSI's effectiveness in forensic document analysis. Following this, Luo et al. [36] proposed an innovative method utilizing anomaly detection to distinguish inks in unbalanced proportions, marking a significant milestone in hyperspectral analysis for ink differentiation. As the field progressed, researchers explored various approaches, including Khan et al. [35], who applied Fuzzy C-Means Clustering in multispectral imaging for enhanced ink discrimination, particularly in scenarios involving mixed inks. Concurrently, Jaiswal et al. [34] introduced deep learning techniques for ink mismatch detection in hyperspectral images, achieving high accuracy through spectral pixel classification.

Recent developments further illustrate the potential of HSI in forgery detection. In 2020, Rahiche and Cheriet [33] employed a graph-regularized orthogonal Nonnegative Matrix Factorization model for unsupervised ink classification in hyperspectral document images, providing a rapid and reliable examination tool that does not require training data. The following year, Jaiswal et al. [31] [32] developed the innovative unsupervised deep learning approach, CAE-LR, which combines Convolutional Autoencoders with Logistic Regression for feature extraction and classification. This method significantly outperformed existing techniques in detecting forged documents with mixed-ink compositions, further enhanced by analyzing spectral, spatial, and combined features [30]. Additionally, Iqbal and Sheikh [29] presented a novel approach utilizing HSI to detect ink mismatches, creating a comprehensive database of handwritten notes and employing spectral clustering to overcome traditional limitations in forensic document examination.

In summary, although hyperspectral and multispectral imaging offer advanced tools for detecting document forgeries, their effectiveness can be affected by ink differences, document condition, and forgery complexity. Limitations include the need for specialized equipment and expertise, along with challenges in result interpretation, which may impact their practical use in forensic work.

4.6. Frequency Domain Analysis

Another analysis in document forgery detection has brought attention to the frequency domain features. Qu et al. [6] present a multi-modality Transformer-based framework that combines visual and frequency features, using discrete cosine transform (DCT) coefficients to detect subtle tampering cues, which enhances feature representation in a Swin-Transformer encoder. Nandanwar et al. [38] employ a Fourier spectrum-based method to analyze brightness distribution and spectrum irregularities, achieving high classification rates for forged text in video and document images. Darem et al. [39] introduce unsupervised algorithms that detect Copy-Move and Copy-Paste forgeries directly in JPEG-compressed images, leveraging DCT coefficients for effective localization without decompression. Together, these studies highlight the diverse methodologies utilizing frequency domain features to improve document forgery detection across various contexts.

Frequency domain analysis improves forgery detection but its effectiveness depends on the forgery type and input image quality. Limitations include sensitivity to noise and compression artifacts, complicating detection and posing challenges for real-time applications.

4.7. Deep Learning-Based Methods

Recently, the deep learning-based forgery detection methods aim to enhance accuracy by employing specialized network architectures to identify forgery features in document images. Each method utilizes feature fusion or attention mechanisms to improve sensitivity to tampered regions. As shown in Table 1, the relevant papers using

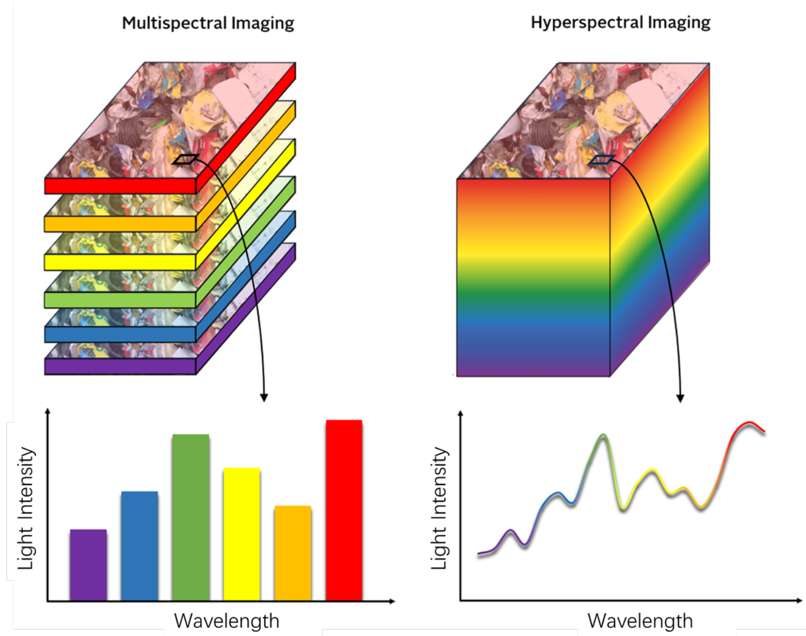

Figure 5. HSI and MSI schematics

deep learning-based methods in document forgery detection are listed. Yang et al. [7] combines CNN with Error Level Analysis (ELA) to detect pixel-level compression inconsistencies, excelling in competitive testing scenarios. Xu et al. [8] employs a dual-stream network that fuses structural and pixel-level features, achieving robust detection under noise and compression. Sun et al. [9] introduces a Multi-Level Feature Attention Network (MFAN) with multi-scale pooling and recalibration, effectively handling variable-scale tampering in sensitive documents. Finally, Liang et al. [40] uses an encoder-decoder network designed to detect image blending, making it particularly suited for real-world forgery scenarios involving complex manipulation layers.

While deep learning-based methods significantly enhance the accuracy of forgery detection, their effectiveness can depend on the quality and diversity of training data. Limitations may include the need for substantial computational resources and the risk of overfitting to specific types of forgeries, which can reduce generalizability to new or unseen manipulation techniques.

5. Conclusion

Document forgery detection is critical for maintaining document integrity as forgery methods become more advanced. This paper reviews detection approaches across several key aspects: the ”Types and Challenges of Document Forgery,” which categorizes tampering techniques and highlights detection obstacles; ”Evaluation Metrics,” outlining measures like IoU, Precision, and F1-score to assess system performance; and the ”Method” section, which examines major detection techniques. Early spatial analysis and printer source-based methods detect common forgeries but struggle with complex cases and may be limited by factors like document quality. Security verification techniques provide authenticity but can require expertise and may be vulnerable to advanced attacks. Visual feature-based methods and spectral imaging capture nuanced inconsistencies but face challenges from noise, equipment needs, and forgery complexity. Frequency domain analysis and deep learning approaches enhance detection precision, yet both depend on data quality and computational resources, with deep learning facing potential overfitting issues.

Table 1. The relevant papers using deep learning-based methods in document forgery detection

Author and Year	Major Findings
Yang P et al., 2022 [7]	This study proposed a document image forgery detection approach based on deep learning, combining Error Level Analysis (ELA) with RGB features to identify and localize forgery in documents. The model achieved competitive performance in the 5th Ali Tianchi forgery detection competition, placing well among 1,470 participating teams, indicating its effectiveness for practical applications.
W. Xu et al., 2022 [8]	This paper presented a dual-stream network for forgery detection in document images, composed of a Spatial Information Extraction Network (SIEN) and Correlation Feature Extraction Network (CFEN). This network effectively identifies structural and pixel-level anomalies, using feature fusion and Xception-based discriminative layers to improve detection accuracy. Its robustness to compression and noise enables reliable forgery localization under diverse conditions.
Sun Y, 2022 [9]	The study introduced the Multi-level Feature Attention Network (MFAN) for detecting forgeries in certificate images. By employing Atrous Spatial Pyramid Pooling and feature recalibration techniques, the network enhances sensitivity to tampered regions and adapts to different manipulation scales. MFAN showed superior performance compared to traditional forensic methods in detecting fake certificate images.
W. Liang, 2022 [40]	This paper proposed a robust encoder-decoder network designed to localize forgeries in document images, particularly those processed with image blending techniques. Through multi-scale feature extraction and attention-based modules, the network enhances its sensitivity to both local and global features, with validation on a custom dataset resembling real-world forgery scenarios demonstrating its reliability.

In summary, the review provides a structured comparison of these methods, offering guidance for developing more adaptable, accurate forgery detection systems to meet evolving security needs.

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